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1 Introduction

This document provides usage information for Physical configuration Audit Script. The script provides a mechanism where presence of and size of files in /system, /data, /etc and /opt are recorded on a test target and can then be checked for their existence and size on another target so that a physical configuration audit can be performed by a Tier1 to detect missing or incorrectly changed files from the test configuration

The target audience for this document are Tier1 to conduct physical configuration audit on Continental software release received. Script and procedure is same for both regular and Sdk release.

1.1 Notational Conventions

Ctrl+x	*A sequence with a '+' indicates that you must press and hold one key or pointing device button while pressing another key or pointing device button*
[ ]	*Optional elements in a command*
…..	*Indicates one of the following possibilities:* • *Additional optional arguments in a statement have been omitted* • *The preceding item or items can be repeated one or more times* • *Additional parameters, values, or other information can be entered*
-	*Line continuation: indicates that a command line or code line continues on the following line when at the end of a line*
Italic text	*Indicates important information, complete titles of manuals or variables*
`Mono-spaced`	*Example code or interactive screen display/output*
Numbers	*All numbers are assumed to be decimal unless otherwise noted. Non decimal radixes – binary, octal or hexadecimal – are explicitly indicated*
Underline blue text	*A link to an external reference (e.g., website, document)*

`Modules`
`  |-> section`
`  |-> item`
	A link to a section or item in the Telematics Platform html interface specification (Doxygen output from platform code headers)

Diagram Legend

- Telematics Platform
- Product Specific
- 3rd Party

2 Setup

2.1 Tools/Equipment Needed

- Continental target that could be flashed, with necessary harness / test fixture.
- 12-volt power supply
- High speed USB cable for access to Android Debug bridge of 5G NAD module
- Personal Computer (PC) with Windows 7 or Windows 10 operating system, note that Windows 10 is recommended (preferred)
- ADB (Refer Documentation here)

2.2 Setup Instructions

1. Install USB host driver compatible with ADB
2. Install Android Debug Bridge (ADB) on PC.
3. Connect NAD module HS USB port to the PC via High Speed USB cable
4. Execute adb shell on windows command prompt and enter into command shell of sa515 NAD target.

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2.3 Setup Troubleshooting

Not Applicable

3 Features/Components

3.1 Physical Configuration Audit

3.1.1 Overview

3.1.1.1 Platform Content Overview

Platform provides a mechanism to detect missing or incorrectly changed files using shell scripts. Sha1sum of all files in /system, /data, /etc and /opt folders on test target are recorded before software release by Continental. The recorded size and sha1sum of all files in text format will be part of the release package. Using a second shell script the recorded sha1sum and size are parsed and compared to the released software flashed on another target to identify the missing and incorrect files.

3.1.1.2 Product Expectations

- Install Android Debug Bridge (ADB) on PC.
- Connect the NAD module HS USB port to the PC via High Speed USB cable

3.1.1.3 Extending Platform Functionality

Not Applicable

3.1.2 Setup

3.1.2.1 Tools/Equipment Needed

Not Applicable

3.1.2.2 Setup Instructions

Not Applicable

3.1.2.3 Setup Troubleshooting

Not Applicable

3.1.3 Usage

3.1.3.1 Basic Instructions

The file pca-audit-file.txt which is part of Continental release package contains recorded sha1sum and size of all files on test target. When the released software is flashed on a different target, the script pca-audit.sh could be run to conduct audit to find out missing and incorrect files with respect to test target. The results will be stored in pca-audit-result.txt in data partition.

3.1.3.2 Detailed Instructions for Continental release process

1. Execute pca-audit-file-create.sh (/usr/bin/pca-audit-file-create.sh) to record sha1sum and size of all files in /system, /data, /etc and /opt in pca-audit-file.txt. pca-audit-file.txt gets created in data partition.
2. Upload pca-audit-file.txt in release/metrics in the released software.

3.1.3.3 Detailed instructions for Tier1

1. Execute adb push pca-audit-file.txt /data.
2. Execute pca-audit.sh (/usr/bin/pca-audit.sh). This script compares sha1sum and size of all files on target to the sha1sum and size of files listed in pca-audit-file.txt. Results are stored in pca-audit-result.txt in data partition.
3. The text file pca-audit-result.txt contains
   a) list of files in pca-audit-file.txt not present on the target.

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b) list of files in pca-audit-file.txt that are bigger than the ones on target

c) list of files in pca-audit-file.txt that are smaller than the ones on target

Note:- The files pca-audit-file.txt, pca-audit-result.txt and /data/log/messages will always get listed in pca-audit-result.txt. These files will be different between the test target and released software. These or any similar files need not be considered incorrect in the released software. Embedded below is an example pca-audit-result.txt.

pca-audit-result.txt

3.1.3.4 Testing/Error Reporting

Not Applicable

4 Known Issues

Not Applicable

5 FAQs

Not Applicable

6 Troubleshooting/Tips and Tricks

Not Applicable

7 References

7.1 Abbreviations/Acronyms/Definitions

Not Applicable

7.2 Platform Documentation

TP_102G-00001: Telematics Platform Glossary

7.3 Website links

N/A

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8 Revision History

Version	Date	Description
01.00	Jul 1, 2020	Initial release

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